The Æther-flow Interpretation of Relativity

Primitive-Reservoir Observer-Reduction
Section-Centered Hidden-Response
Projection-Operator Core
Collapse to a Canonical Minimal
Visible-Response Representative

Abstract

The current active-primary theorem already proves that, once the fixed full canonical current-body image core is held fixed, every benchmark-faithful reduced projection-response substrate structure compatible with that core induces the same visible quadratic response operator on the fixed section-centered displacement fibers and differs from the canonical minimal representative only by inert kernel extension. The current routing criterion therefore allows two equivalent honest next moves below that frontier: derive the reduced projection-response substrate structure itself from still more primitive explicit substrate structure, or derive its canonical minimal visible-response representative.

The present note takes the second route in the cleanest recorded older package still available. Starting from a benchmark-faithful current-body-realizing section-centered hidden-response projection-operator core $\mathfrak{S}^{\mathrm{disp,resp,projop}}$, it proves that the only benchmark-facing datum still carried there is the positive visible-response operator

$$S_\sigma^{\mathrm{disp,resp,vis}}(y) = \left(P_\sigma^{\mathrm{disp,resp}}(y) K_\sigma^{\mathrm{disp,resp}}(y)^{-1} P_\sigma^{\mathrm{disp,resp}}(y)^*\right)^{-1}$$

on the fixed displacement fibers. Equivalently, the full projection-operator-core presentation collapses benchmark-facingly to the canonical minimal visible-response representative

$$\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}} = \left(V_\sigma^{\mathrm{pppresec}}, \mathrm{id}, S_\sigma^{\mathrm{disp,resp,vis}}\right)$$

over the fixed section-centered displacement body. The larger carrier, its full substrate-body presentation, and all retained kernel directions contribute only benchmark-neutral excess structure.

The benchmark boundary remains fixed throughout. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, the low-energy matter law is not enlarged, and no stronger promotion language is licensed. The positive result is therefore a genuine deeper derivation in the explicit older package named by the routing layer, without inventing another same-output presentation. The next honest burden moves below the projection-operator core and below carrier-level reduced-structure presentation to the visible-response representative itself.

1. Introduction

The active derivational program of Æther-flow Interpretation of Relativity has reached a narrower and more disciplined stage on the section-centered hidden-response line. The earlier projection-operator-

core collapse theorem already showed that the full displacement-response substrate law carries no live benchmark-facing freedom beyond a smaller current-body-realizing projection-operator core. The later current-body-image-core theorem then showed that, once a reduced projection-response substrate structure is fixed, the whole canonical current-body image core is already a canonical and unique output. The most recent reduced-structure necessity / uniqueness theorem sharpened that gain further: once the fixed current image core is held fixed, every compatible reduced structure carries the same visible quadratic response operator and differs only by inert kernel extension.

At that point the routing layer named the next honest move clearly. One could either derive the reduced projection-response substrate structure itself from still more primitive explicit substrate structure, or equivalently derive its canonical minimal visible-response representative. The present note takes the second route, using the cleanest recorded older explicit package still available without inventing a new same-output carrier presentation.

The key observation is that the current-body-realizing projection-operator core already determines a unique positive visible quadratic response on the fixed displacement fibers. Once that operator is read off, the full substrate-body presentation and any residual carrier/kernel directions cease to be benchmark-facingly live. They contribute at most nonnegative excess energy above the same visible displacement datum and do not alter the fixed current body-response core or any downstream benchmark-facing consequence.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays strictly inside that boundary. It does not introduce a second operative metric, it does not enlarge the low-energy matter law, and it does not claim a completed first-principles derivation of GR from final substrate microphysics. Its narrower benchmark-facing role is to show that the older projection-operator-core presentation already collapses to the visible-response representative on the fixed section-centered displacement fibers.

2. Fixed Inputs

Definition 1 (Recorded primitive continuation data)**.** *Fix the following continuation data from the active primitive-reservoir observer-reduction line.*

1. *For each sector label* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$*, a primitive forcing shell*

$$\mathcal{I}_\sigma^{\mathrm{for}}.$$

2. *The product shell*

$$\mathcal{S}_{\mathrm{prim}} \equiv \mathcal{I}_{\mathrm{car}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{cur}}^{\mathrm{for}} \times \mathcal{I}_{\mathrm{hom}}^{\mathrm{for}}.$$

3. *The recorded metric and ordinary-matter outputs*

$$\mathscr{G}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \to \{\textit{observer metrics}\}, \qquad \mathscr{T}_{\mathrm{out}} : \mathcal{S}_{\mathrm{prim}} \times \Psi \to \{\textit{observer stress tensors}\},$$

satisfying

$$\mathscr{G}_{\mathrm{out}}(\mathbf{w}) = g^{\sharp}, \qquad \mathscr{T}_{\mathrm{out}}(\mathbf{w}, \psi) = T^{\mathrm{matter}}_{\mu\nu}[g^{\sharp}, \psi] \tag{1}$$

for every $\mathbf{w} \in \mathcal{S}_{\mathrm{prim}}$ *and every matter configuration* $\psi \in \Psi$.

Definition 2 (Recorded benchmark base and downstream chain)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *The active line already fixes the following data and consequences.*

1. *the current proto-section benchmark base, in particular the proto-section body* $\mathcal{B}^{\mathrm{psec}}_{\sigma}$ *and shell* $\mathcal{Z}^{\mathrm{psec}}_{\sigma}$*;*

2. *the current section-centered displacement body*

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma} \subset \mathcal{B}^{\mathrm{psec}}_{\sigma} \times V^{\mathrm{pppresec}}_{\sigma}$$

whose fiber

$$\mathcal{D}^{\mathrm{pppresec}}_{\sigma}(y) \equiv \{v \in V^{\mathrm{pppresec}}_{\sigma} : (y, v) \in \mathcal{D}^{\mathrm{pppresec}}_{\sigma}\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}^{\mathrm{psec}}_{\sigma}$*;*

3. *the recorded downstream fact that, once a benchmark-faithful quadratic body-response route on the fixed displacement body is available, the packages*

$$\begin{aligned}
&\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyresp}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,bodyop}}, \\
&\mathfrak{S}^{\mathrm{sub,proto,pppresec,pb,op}}, \quad \mathfrak{S}^{\mathrm{sub,proto,pppresec,pb}}, \\
&\mathfrak{R}^{\mathrm{sub,proto,presec,res}}, \quad \mathfrak{R}^{\mathrm{sub,proto,presec}}, \\
&\mathfrak{R}^{\mathrm{sub,proto,presec,sec}}
\end{aligned}$$

together with the previously recorded continuity-Hessian compressed core and benchmark one-metric observer package follow unchanged;

4. *benchmark faithfulness, meaning preservation of* (1)*, no second operative metric, no enlarged low-energy matter law, no change to the ordinary matter route, and no premature promotion from adoption to completed derivation.*

Definition 3 (Current-body-realizing section-centered displacement-response projection-operator core)**.** *Fix* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$. *A* benchmark-faithful current-body-realizing section-centered displacement-response projection-operator core $\mathfrak{S}^{\mathrm{disp,resp,projop}}$ *for sector* σ *consists of:*

1. *a finite-dimensional Euclidean substrate carrier*

$$U^{\mathrm{disp,resp}}_{\sigma};$$

2. *a compact convex section-centered substrate body*

$$\mathcal{U}^{\mathrm{disp,resp}}_{\sigma} \subset \mathcal{B}^{\mathrm{psec}}_{\sigma} \times U^{\mathrm{disp,resp}}_{\sigma}$$

whose fiber

$$\mathcal{U}_\sigma^{\mathrm{disp,resp}}(y) \equiv \left\{ u \in U_\sigma^{\mathrm{disp,resp}} : (y,u) \in \mathcal{U}_\sigma^{\mathrm{disp,resp}} \right\}$$

is nonempty, compact, convex, contains 0*, and has nonempty interior for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*;*

3. *a smooth family of surjective linear displacement projections*

$$P_\sigma^{\mathrm{disp,resp}}(y) : U_\sigma^{\mathrm{disp,resp}} \to V_\sigma^{\mathrm{ppprese c}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}};$$

4. *a smooth family of positive self-adjoint substrate-response operators*

$$K_\sigma^{\mathrm{disp,resp}}(y) : U_\sigma^{\mathrm{disp,resp}} \to U_\sigma^{\mathrm{disp,resp}}, \qquad y \in \mathcal{B}_\sigma^{\mathrm{psec}},$$

satisfying the uniform coercive bound

$$\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y) u \right\rangle \geq \mu_\sigma \|u\|^2 \qquad \text{for all } y \in \mathcal{B}_\sigma^{\mathrm{psec}}, \quad u \in U_\sigma^{\mathrm{disp,resp}} \tag{2}$$

for some constant $\mu_\sigma > 0$*;*

5. *exact-shell discipline:*

$$\left\{ y \in \mathcal{B}_\sigma^{\mathrm{psec}} : \exists u \in \mathcal{U}_\sigma^{\mathrm{disp,resp}}(y) \text{ with } P_\sigma^{\mathrm{disp,resp}}(y) u = 0 \right\} = \mathcal{Z}_\sigma^{\mathrm{psec}};$$

6. *current-body realization:*

$$\mathcal{D}_\sigma^{\mathrm{ppprese c}}(y) = P_\sigma^{\mathrm{disp,resp}}(y)(\mathcal{U}_\sigma^{\mathrm{disp,resp}}(y)) \qquad \text{for every } y \in \mathcal{B}_\sigma^{\mathrm{psec}}; \tag{3}$$

7. *benchmark faithfulness in the sense of Definition 2(4).*

Remark 1. *Definition 3 is an older explicit package now preserved on the active line. The present note does not restore it as the live burden. It asks the sharper question forced by the current criterion: once the active line already knows that reduced-structure carrier extensions are benchmark neutral, does any benchmark-facing datum remain in the full projection-operator-core presentation beyond the visible quadratic response on the fixed displacement fibers?*

3. Canonical Visible-Response Representative

Proposition 1 (Positive Schur-response operator)**.** *Fix a benchmark-faithful current-body-realizing projection-operator core in the sense of Definition 3. For each* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*, define*

$$M_\sigma^{\mathrm{disp,resp}}(y) \equiv P_\sigma^{\mathrm{disp,resp}}(y) K_\sigma^{\mathrm{disp,resp}}(y)^{-1} P_\sigma^{\mathrm{disp,resp}}(y)^* : V_\sigma^{\mathrm{ppprese c}} \to V_\sigma^{\mathrm{ppprese c}},$$

where $(\cdot)^*$ *denotes Euclidean adjoint with respect to the fixed inner products on* $U_\sigma^{\mathrm{disp,resp}}$ *and* $V_\sigma^{\mathrm{ppprese c}}$*. Then* $M_\sigma^{\mathrm{disp,resp}}(y)$ *is positive self-adjoint and invertible for every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$*.*

Proof. Self-adjointness is immediate from the self-adjointness of $K_\sigma^{\mathrm{disp,resp}}(y)^{-1}$. For any $0 \neq w \in V_\sigma^{\mathrm{ppprese c}}$,

$$\left\langle w, M_\sigma^{\mathrm{disp,resp}}(y) w \right\rangle = \left\langle K_\sigma^{\mathrm{disp,resp}}(y)^{-1/2} P_\sigma^{\mathrm{disp,resp}}(y)^* w, K_\sigma^{\mathrm{disp,resp}}(y)^{-1/2} P_\sigma^{\mathrm{disp,resp}}(y)^* w \right\rangle > 0$$

because $P_\sigma^{\mathrm{disp,resp}}(y)$ is surjective and hence $P_\sigma^{\mathrm{disp,resp}}(y)^*$ is injective. Thus $M_\sigma^{\mathrm{disp,resp}}(y)$ is positive definite and invertible on the finite-dimensional space $V_\sigma^{\mathrm{pppresec}}$. □

Definition 4 (Canonical lift and visible-response operator)**.** *Fix a benchmark-faithful current-body-realizing projection-operator core. For each $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$, define the* visible-response operator

$$S_\sigma^{\mathrm{disp,resp,vis}}(y) \equiv M_\sigma^{\mathrm{disp,resp}}(y)^{-1} : V_\sigma^{\mathrm{pppresec}} \to V_\sigma^{\mathrm{pppresec}} \tag{4}$$

and the canonical visible lift

$$J_\sigma^{\mathrm{disp,resp,can}}(y) \equiv K_\sigma^{\mathrm{disp,resp}}(y)^{-1} P_\sigma^{\mathrm{disp,resp}}(y)^* S_\sigma^{\mathrm{disp,resp,vis}}(y) : V_\sigma^{\mathrm{pppresec}} \to U_\sigma^{\mathrm{disp,resp}}. \tag{5}$$

Define the induced visible quadratic response on the fixed displacement body by

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y,v) \equiv \frac{1}{2}\left\langle v, S_\sigma^{\mathrm{disp,resp,vis}}(y)v\right\rangle \qquad \text{for } (y,v) \in \mathcal{D}_\sigma^{\mathrm{pppresec}}. \tag{6}$$

Proposition 2 (Canonical lift and energy decomposition)**.** *Fix a benchmark-faithful current-body-realizing projection-operator core. Then, for every $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$,*

1. $$P_\sigma^{\mathrm{disp,resp}}(y) \circ J_\sigma^{\mathrm{disp,resp,can}}(y) = \mathrm{id}_{V_\sigma^{\mathrm{pppresec}}};$$

2. $$\left\langle J_\sigma^{\mathrm{disp,resp,can}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)k\right\rangle = 0$$

   for every $v \in V_\sigma^{\mathrm{pppresec}}$ and every $k \in N_\sigma^{\mathrm{disp,resp}}(y) \equiv \ker P_\sigma^{\mathrm{disp,resp}}(y)$;

3. *every $u \in U_\sigma^{\mathrm{disp,resp}}$ admits the unique decomposition*

   $$u = J_\sigma^{\mathrm{disp,resp,can}}(y)(P_\sigma^{\mathrm{disp,resp}}(y)u) + k, \qquad k \in N_\sigma^{\mathrm{disp,resp}}(y);$$

4. *for $u \in U_\sigma^{\mathrm{disp,resp}}$, with*

   $$v \equiv P_\sigma^{\mathrm{disp,resp}}(y)u, \qquad k \equiv u - J_\sigma^{\mathrm{disp,resp,can}}(y)v,$$

   the energy splits as

   $$\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u\right\rangle = \left\langle v, S_\sigma^{\mathrm{disp,resp,vis}}(y)v\right\rangle + \left\langle k, K_\sigma^{\mathrm{disp,resp}}(y)k\right\rangle. \tag{7}$$

Proof. Item (1) follows directly from Definitions 4 and 1:

$$P_\sigma^{\mathrm{disp,resp}}(y) J_\sigma^{\mathrm{disp,resp,can}}(y) = M_\sigma^{\mathrm{disp,resp}}(y) S_\sigma^{\mathrm{disp,resp,vis}}(y) = \mathrm{id}.$$

For item (2), let $k \in N_\sigma^{\mathrm{disp,resp}}(y)$. Then

$$\begin{aligned}\left\langle J_\sigma^{\mathrm{disp,resp,can}}(y)v, K_\sigma^{\mathrm{disp,resp}}(y)k\right\rangle &= \left\langle P_\sigma^{\mathrm{disp,resp}}(y)^* S_\sigma^{\mathrm{disp,resp,vis}}(y)v, k\right\rangle \\ &= \left\langle S_\sigma^{\mathrm{disp,resp,vis}}(y)v, P_\sigma^{\mathrm{disp,resp}}(y)k\right\rangle \\ &= 0.\end{aligned}$$

For item (3), if $u \in U_\sigma^{\mathrm{disp,resp}}$, set

$$k \equiv u - J_\sigma^{\mathrm{disp,resp,can}}(y)\big(P_\sigma^{\mathrm{disp,resp}}(y)u\big).$$

Applying $P_\sigma^{\mathrm{disp,resp}}(y)$ and using item (1) gives $P_\sigma^{\mathrm{disp,resp}}(y)k = 0$, so $k \in N_\sigma^{\mathrm{disp,resp}}(y)$. Uniqueness follows because item (1) fixes the non-kernel part. Finally, item (4) is obtained by expanding

$$u = J_\sigma^{\mathrm{disp,resp,can}}(y)v + k$$

and using item (2). □

Proposition 3 (Visible response is exactly the benchmark-facing quadratic datum)**.** *Fix a benchmark-faithful current-body-realizing projection-operator core. Then:*

1. *the induced response $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}$ of (6) is smooth in y, fiberwise quadratic in v, and satisfies*

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y,v) = 0 \qquad \Longleftrightarrow \qquad v = 0;$$

2. *the exact-shell relation on the fixed displacement body is*

$$\{y \in \mathcal{B}_\sigma^{\mathrm{psec}} : 0 \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}(y)\} = \mathcal{Z}_\sigma^{\mathrm{psec}};$$

3. *for every $u \in U_\sigma^{\mathrm{disp,resp}}$ with $v = P_\sigma^{\mathrm{disp,resp}}(y)u$,*

$$\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y,v) \le \frac{1}{2}\left\langle u, K_\sigma^{\mathrm{disp,resp}}(y)u\right\rangle,$$

with equality if and only if $u = J_\sigma^{\mathrm{disp,resp,can}}(y)v$.

Proof. Item (1) follows because $S_\sigma^{\mathrm{disp,resp,vis}}(y)$ is a smooth positive self-adjoint operator family, so $v \mapsto \frac{1}{2}\left\langle v, S_\sigma^{\mathrm{disp,resp,vis}}(y)v\right\rangle$ is quadratic and positive definite on each fiber. For item (2), if $0 \in \mathcal{D}_\sigma^{\mathrm{ppprese c}}(y)$, then the current-body realization (3) gives $u \in \mathcal{U}_\sigma^{\mathrm{disp,resp}}(y)$ with $P_\sigma^{\mathrm{disp,resp}}(y)u = 0$, so Definition 3(5) yields $y \in \mathcal{Z}_\sigma^{\mathrm{psec}}$. The converse is immediate from the same two conditions. Item (3) is exactly the energy split (7). □

Definition 5 (Canonical minimal visible-response representative)**.** *For sector σ, define the* canonical minimal visible-response representative

$$\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}} \equiv \left(V_\sigma^{\mathrm{ppprese c}}, \mathrm{id}_{V_\sigma^{\mathrm{ppprese c}}}, S_\sigma^{\mathrm{disp,resp,vis}}\right)$$

over the fixed displacement body $\mathcal{D}_\sigma^{\mathrm{ppprese c}}$, the exact-shell relation of Proposition 3(2), and the benchmark-faithfulness conditions of Definition 2(4).

Remark 2. *Definition 5 keeps only what remains visible below the older projection-operator-core frontier: the fixed section-centered displacement fibers together with their positive quadratic response operator. No ambient carrier choice, substrate-body presentation, or kernel-extension datum is retained there as separately primitive data.*

4. Benchmark-Neutrality of the Larger Presentation

Proposition 4 (Substrate-body and kernel directions beyond the visible representative are benchmark neutral)**.** *Fix a benchmark-faithful current-body-realizing projection-operator core. For every* $y \in \mathcal{B}_\sigma^{\mathrm{psec}}$ *and every* $u \in \mathcal{U}_\sigma^{\mathrm{disp,resp}}(y)$*, let*

$$v \equiv P_\sigma^{\mathrm{disp,resp}}(y)u \in \mathcal{D}_\sigma^{\mathrm{ppresec}}(y), \qquad k \equiv u - J_\sigma^{\mathrm{disp,resp,can}}(y)v.$$

Then:

1. $k \in N_\sigma^{\mathrm{disp,resp}}(y)$*;*
2. *u and the canonical visible representative $J_\sigma^{\mathrm{disp,resp,can}}(y)v$ project to the same displacement datum v;*
3. *the excess contribution of u beyond the canonical visible representative is exactly the nonnegative kernel term*
$$\frac{1}{2}\left\langle k, K_\sigma^{\mathrm{disp,resp}}(y)k\right\rangle;$$
4. *varying only the kernel part k changes no visible response datum $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y,v)$ on the fixed displacement body.*

Proof. Items (1) and (2) are Proposition 2(3). Item (3) is Proposition 2(4). Item (4) follows because $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}(y,v)$ depends only on v and $S_\sigma^{\mathrm{disp,resp,vis}}(y)$, both unchanged under variation of k. □

Proposition 5 (The downstream benchmark-facing package already factors through the minimal representative)**.** *For every benchmark-faithful current-body-realizing projection-operator core, the pair*

$$\left(\mathcal{D}_\sigma^{\mathrm{ppresec}}, \widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}\right)$$

is a benchmark-faithful quadratic body-response route on the fixed displacement body, so the downstream package recorded in Definition 2(3) follows unchanged from the canonical minimal visible-response representative $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$*.*

Proof. Proposition 3(1) gives the required quadratic-response properties on the fixed displacement body, and Proposition 3(2) gives the exact-shell relation. Benchmark faithfulness is inherited from Definition 3(7). The downstream fact of Definition 2(3) therefore applies directly to the body-response route defined by $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}$. □

Proposition 6 (The benchmark package remains fixed)**.** *The canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *preserves the same benchmark one-metric package* (1)*. In particular, the operative metric remains exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged.*

Proof. This is exactly the retained benchmark-faithfulness condition in Definition 5, inherited from Definition 3(7). □

5. Main Theorem

Theorem 1 (Projection-operator core collapse to a canonical minimal visible-response representative)**.** *Fix the primitive continuation data of Definition 1, the recorded benchmark base and downstream chain of Definition 2, and for each sector* $\sigma \in \{\mathrm{car}, \mathrm{cur}, \mathrm{hom}\}$ *a benchmark-faithful current-body-realizing projection-operator core in the sense of Definition 3. Then the older projection-operator-core frontier already collapses benchmark-facingly to the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$*. More precisely:*

1. *the projection-operator core canonically determines the positive visible-response operator* $S_\sigma^{\mathrm{disp,resp,vis}}$ *and the induced quadratic response* $\widehat{\mathscr{R}}_\sigma^{\mathrm{disp,resp,vis}}$ *on the fixed displacement body by Definition 4;*

2. *the fixed displacement body together with* $S_\sigma^{\mathrm{disp,resp,vis}}$ *defines a benchmark-faithful canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *in the sense of Definition 5;*

3. *every point of the full substrate carrier, and in particular every point of the recorded section-centered substrate body, decomposes uniquely into the canonical visible representative of the same displacement datum together with a kernel direction, and the excess contribution beyond the visible representative is exactly the nonnegative kernel-energy term of Proposition 2(4);*

4. *the full substrate-body presentation, all retained carrier/kernel directions, and every same-visible-response reduced carrier presentation are benchmark neutral at current scope because the downstream benchmark-facing package already factors through* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *by Proposition 5;*

5. *the benchmark boundary remains fixed by Proposition 6;*

6. *consequently the next honest burden below this frontier is no longer another replay of the projection-operator core, another restored reduced carrier presentation, or another inert-kernel extension relay. It is to derive or justify the canonical minimal visible-response representative* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$*, equivalently the visible-response operator* $S_\sigma^{\mathrm{disp,resp,vis}}$ *on the fixed section-centered displacement fibers, from still more primitive explicit substrate structure, or to prove a still sharper theorem strictly inside that visible representative.*

Proof. Item (1) is Definition 4 together with Proposition 1. Item (2) is Definition 5 and Proposition 3. Item (3) is Proposition 2, with the benchmark-neutral interpretation spelled out in Proposition 4. Item (4) is exactly Proposition 5. Item (5) is Proposition 6. Item (6) is the routing consequence of items (1)–(5): once the visible-response representative is already forced from the older explicit projection-operator-core package, another manuscript that merely re-presents the same visible operator through a larger carrier or body or inert kernel extension no longer removes any remaining benchmark-facing freedom. □

Corollary 1 (Primary post-collapse burden)**.** *After Theorem 1, the next honest benchmark-facing manuscript below the present frontier must do at least one of the following:*

1. *derive or force the visible-response operator* $S_\sigma^{\mathrm{disp,resp,vis}}$ *itself on the fixed displacement fibers from still more primitive explicit substrate structure;*

2. *equivalently derive* $\mathfrak{S}_\sigma^{\mathrm{disp,resp,red,min}}$ *from still more primitive explicit substrate structure;*

3. *prove a sharper necessity, uniqueness, or compatibility theorem strictly inside that visible representative while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary.*

Another manuscript that merely re-adjoins kernel directions, re-expands to a larger reduced carrier presentation, replays the projection-operator core, returns to the full displacement-response substrate law, or re-expands back to the larger pullback-body-operator, pullback-operator, pullback-quadratic, hidden-response, residual-normal-form, or pre-proto-section presentations is not the honest default next move.

Proof. Theorem 1 shows that the visible-response representative is already the remaining benchmark-facing datum below the older projection-operator-core frontier. Therefore a subsequent manuscript must improve on that visible datum rather than merely accompany it through a larger same-output presentation. □

6. Conclusion

The positive result of this note is a genuine benchmark-facing collapse below the older projection-operator-core frontier. The full current-body-realizing projection-operator core is no longer the honest stopping point even at carrier scope. Its benchmark-facing content already reduces to the fixed section-centered displacement body together with the positive visible-response operator read off from the Schur complement formula. Equivalently, the active line now canonically extracts the minimal visible-response representative $\mathfrak{S}^{\mathrm{disp,resp,red,min}}$ from that older explicit package.

The scope remains disciplined. The one operative metric is still exactly $g^\sharp$, the ordinary matter route is unchanged, there is no second operative metric, and the low-energy matter law is not enlarged. Nothing here upgrades adoption to completed derivation or licenses stronger promotion language. The gain is narrower and more honest than that: it shows that the larger projection/body/carrier presentation is benchmark-neutral once the visible quadratic response on the fixed displacement fibers has been identified.

The next open burden is therefore clear. The next benchmark-facing manuscript should derive or justify the visible-response representative itself from still more primitive explicit substrate structure, equivalently derive the positive visible-response operator on the fixed section-centered displacement fibers, while preserving the same benchmark one-metric package, the same ordinary matter route, and the same claim boundary. Another same-output kernel-extension relay, another restored reduced carrier presentation, another replay of the projection-operator core, or another return to the full displacement-response substrate law is not the clean next manuscript target at current scope.